

## 30.3 Group penalties

A factor with  $k$  levels enters a linear model as  $k - 1$  indicator columns (Chapter 15), and a smooth effect as several basis columns. The question is whether *the factor* matters. The lasso on the indicator columns can keep some levels and drop others, and its answer depends on the coding. Yuan and Lin (2006) proposed a penalty that treats a block of columns as a unit.

### 30.3.1 The group lasso

Partition the columns of  $\mathbf{X}$  into groups  $g = 1, \dots, G$ , with  $\mathbf{X}_g$  the  $n \times p_g$  block of group  $g$  and  $\mathbf{b}_g$  the matching block of coefficients, so that  $\mathbf{X}\mathbf{b} = \sum_g \mathbf{X}_g \mathbf{b}_g$ .

#### Definition 30.3.1 — Group lasso

For weights  $w_g > 0$ , the **group lasso** criterion is

$$Q(\mathbf{b}) = \frac{1}{2} \left\| \mathbf{y} - \sum_{g=1}^G \mathbf{X}_g \mathbf{b}_g \right\|^2 + \lambda \sum_{g=1}^G w_g \|\mathbf{b}_g\|, \quad (30.3.1)$$

with Euclidean norms  $\|\mathbf{b}_g\|$ , not squared. The usual weights are  $w_g = \sqrt{p_g}$ . The columns of each group are usually centred and **orthonormalized within the group** before fitting, so that  $\mathbf{X}_g^\top \mathbf{X}_g = \mathbf{I}_{p_g}$ . Different groups are not made orthogonal to each other.

With singleton groups and unit weights this is the lasso. In general it acts like the lasso *across* groups and like ridge *within* them:  $\|\mathbf{b}_g\|$  has a corner only at  $\mathbf{b}_g = \mathbf{0}$ , so a group is either absent or present with, typically, all its coefficients nonzero.

#### Proposition 30.3.1 — Group lasso and block soft thresholding

- (a) A minimizer of Eq. 30.3.1 exists, and all minimizers have the same fitted vector.
- (b) (*KKT conditions.*)  $\hat{\boldsymbol{\beta}}$  is a minimizer iff, with  $\mathbf{r} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$ , for every group  $g$ ,

$$\mathbf{X}_g^\top \mathbf{r} = \lambda w_g \frac{\hat{\boldsymbol{\beta}}_g}{\|\hat{\boldsymbol{\beta}}_g\|} \quad \text{if } \hat{\boldsymbol{\beta}}_g \neq \mathbf{0}, \quad \|\mathbf{X}_g^\top \mathbf{r}\| \leq \lambda w_g \quad \text{if } \hat{\boldsymbol{\beta}}_g = \mathbf{0}. \quad (30.3.2)$$

- (c) Let  $\lambda > 0$ . Then  $\hat{\boldsymbol{\beta}} = \mathbf{0}$  iff  $\lambda \geq \lambda_{\max} = \max_g \|\mathbf{X}_g^\top \mathbf{y}\| / w_g$ .
- (d) (*Orthonormal groups.*) If  $\mathbf{X}_g^\top \mathbf{X}_h = \mathbf{I}_{p_g}$  for  $g = h$  and  $\mathbf{0}$  for  $g \neq h$ , the unique minimizer is **block soft thresholding** of  $\mathbf{z}_g = \mathbf{X}_g^\top \mathbf{y}$ :

$$\hat{\boldsymbol{\beta}}_g = \left( 1 - \frac{\lambda w_g}{\|\mathbf{z}_g\|} \right)_+ \mathbf{z}_g. \quad (30.3.3)$$

- (e) (*Block update; compare Proposition 30.2.2.*) If  $\mathbf{X}_g^\top \mathbf{X}_g = \mathbf{I}_{p_g}$ , then minimizing Eq. 30.3.1 over  $\mathbf{b}_g$  with the other blocks fixed gives Eq. 30.3.3 with  $\mathbf{z}_g$  replaced by

$\mathbf{s}_g = \mathbf{X}_g^\top \mathbf{r}_{(g)}$ , where  $\mathbf{r}_{(g)} = \mathbf{y} - \sum_{h \neq g} \mathbf{X}_h \mathbf{b}_h$  is the partial residual.

*Proof.*

- (a) The penalty is convex and continuous. Since  $\sum_g \|\mathbf{b}_g\| \geq \|\mathbf{b}\|$ , it is at least  $(\min_g w_g) \|\mathbf{b}\|$ . Apply Theorem 30.1.1(a)(ii) and (b).
- (b) Combine Lemma 30.1.2 with Lemma 30.1.3, using the groups as the blocks.
- (c) At  $\hat{\boldsymbol{\beta}} = \mathbf{0}$ , the residual is  $\mathbf{y}$ , and Eq. 30.3.2 says that  $\|\mathbf{X}_g^\top \mathbf{y}\| \leq \lambda w_g$  for every  $g$ . If this holds,  $\mathbf{0}$  is a minimizer. Every minimizer then has fitted vector  $\mathbf{0}$  and, by Theorem 30.1.1(b), penalty 0, so it is  $\mathbf{0}$ . If it fails,  $\mathbf{0}$  is not a minimizer.
- (d) Under the assumption,  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y}\|^2 - 2 \sum_g \mathbf{z}_g^\top \mathbf{b}_g + \sum_g \|\mathbf{b}_g\|^2$ . So Eq. 30.3.1 separates into the problems  $\min_{\mathbf{b}_g} \{\frac{1}{2} \|\mathbf{z}_g - \mathbf{b}_g\|^2 + \lambda w_g \|\mathbf{b}_g\|\}$ , each strictly convex. Let  $c = \lambda w_g$  and let  $\mathbf{b}_g$  be given by Eq. 30.3.3. If  $\|\mathbf{z}_g\| \leq c$ , then  $\mathbf{b}_g = \mathbf{0}$  and  $\|\mathbf{z}_g - \mathbf{0}\| \leq c$ . If  $\|\mathbf{z}_g\| > c$ , then  $\mathbf{b}_g = (\|\mathbf{z}_g\| - c) \mathbf{z}_g / \|\mathbf{z}_g\| \neq \mathbf{0}$  and  $\mathbf{z}_g - \mathbf{b}_g = c \mathbf{z}_g / \|\mathbf{z}_g\| = c \mathbf{b}_g / \|\mathbf{b}_g\|$ . In both cases the one-group version of Eq. 30.3.2 holds.
- (e) As a function of  $\mathbf{b}_g$ , the criterion is  $\frac{1}{2} \|\mathbf{r}_{(g)} - \mathbf{X}_g \mathbf{b}_g\|^2 + \lambda w_g \|\mathbf{b}_g\|$  plus a constant. With  $\mathbf{X}_g^\top \mathbf{X}_g = \mathbf{I}$ , this is the one-group problem of (d) with  $\mathbf{z}_g$  replaced by  $\mathbf{s}_g$ .

■

Block soft thresholding keeps the direction of the least squares vector  $\mathbf{z}_g$  and reduces its length by  $\lambda w_g$ : the lasso rule applied to a length. The weights come from (c): for pure noise and orthonormal groups,  $\|\mathbf{X}_g^\top \mathbf{y}\|^2 / \sigma^2 \sim \chi^2(p_g)$  has mean  $p_g$ , so without  $\sqrt{p_g}$  large groups would enter merely for being large (Exercise (A1)). Block coordinate descent with update (e) has every limit point a minimizer because the penalty is separable across blocks (Tseng, 2001), so it converges when the minimizer is unique; without orthonormality the block problem has no closed form (Exercise (C1)).

### 30.3.2 Factors as groups

Orthonormalizing within groups also makes the group lasso independent of how a factor is coded.

#### Proposition 30.3.2 — Invariance to the coding of a group

- (a) Let  $\mathbf{U}_g$  be  $p_g \times p_g$  orthogonal matrices, and replace every  $\mathbf{X}_g$  by  $\mathbf{X}_g \mathbf{U}_g$ . Then  $\hat{\boldsymbol{\beta}}$  minimizes the original criterion iff the vector with blocks  $\mathbf{U}_g^\top \hat{\boldsymbol{\beta}}_g$  minimizes the new one. The fitted contributions  $\mathbf{X}_g \hat{\boldsymbol{\beta}}_g$  are the same.
- (b) Suppose each group is orthonormalized within itself, so that  $\mathbf{X}_g$  is an orthonormal basis of a subspace  $\mathcal{V}_g$ . Then the minimizers, expressed through the fitted contributions  $\mathbf{X}_g \hat{\boldsymbol{\beta}}_g \in \mathcal{V}_g$ , depend on the groups only through the subspaces  $\mathcal{V}_g$ . In particular, for a factor they do not depend on the choice of reference level or of contrasts.

*Proof.*

- (a) With  $\mathbf{b}_g = \mathbf{U}_g \mathbf{c}_g$ , we have  $\mathbf{X}_g \mathbf{U}_g \mathbf{c}_g = \mathbf{X}_g \mathbf{b}_g$  and  $\|\mathbf{c}_g\| = \|\mathbf{b}_g\|$ . So the new criterion

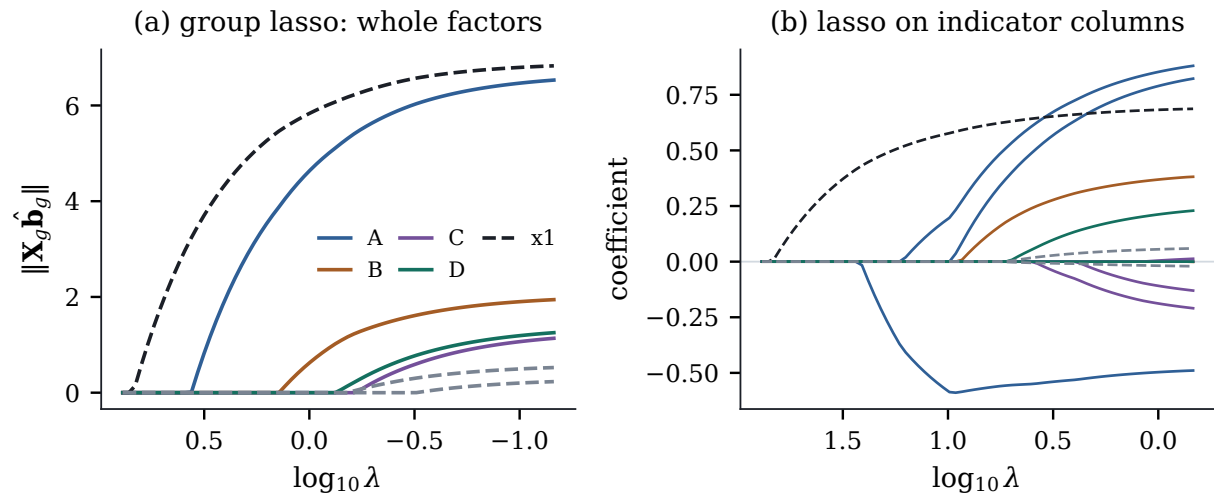


Figure 30.3.1: (a) Group lasso path for [Example \(Selecting factors\)](#): the length of each group's fitted contribution  $\|\mathbf{X}_g \hat{\mathbf{b}}_g\|$  against  $\log_{10} \lambda$ . Factors enter as wholes. (b) The lasso path for the same data, with each indicator column penalized separately, coloured by factor. The dashed curves are the numeric regressors.

at  $\mathbf{c}$  equals the old criterion at  $\mathbf{b}$ , and  $\mathbf{b} \leftrightarrow \mathbf{c}$  is a bijection.

- (b) Two orthonormal bases  $\mathbf{X}_g$  and  $\mathbf{X}'_g$  of  $\mathcal{V}_g$  satisfy  $\mathbf{X}'_g = \mathbf{X}_g \mathbf{U}_g$  with  $\mathbf{U}_g = \mathbf{X}_g^\top \mathbf{X}'_g$  orthogonal. Apply (a). For a factor with  $k$  levels, the centred indicator columns of any reference coding, and the centred columns of any other full-rank contrast coding, span the same  $(k-1)$ -dimensional subspace: the vectors that are constant within levels and orthogonal to  $\mathbf{1}$ .

■

With orthonormalized groups  $\|\mathbf{b}_g\| = \|\mathbf{X}_g \mathbf{b}_g\|$ : the penalty is the total length of the groups' fitted contributions, a coding-free quantity. Simon and Tibshirani (2012) argue for this version.

### Example. (Selecting factors)

The script simulates  $n = 120$  observations of four factors  $A, B, C, D$  (with 4, 3, 5 and 3 levels) and three numeric regressors; the response depends on  $A$ , weakly on  $B$ , and on  $x_1$ . The group lasso path starts at  $\lambda_{\max} = 6.876$ , and groups enter in the order  $x_1, A, B, D$ . At  $\lambda = 2.0$  only  $A$  and  $x_1$  are selected, with  $\|\mathbf{X}_A \hat{\mathbf{b}}_A\| = 2.834$  and  $\|\mathbf{X}_{x_1} \hat{\mathbf{b}}_{x_1}\| = 4.861$ . Another reference level gives the same fitted contributions to machine precision. The lasso on the raw indicator columns does not: at  $\lambda = 6.0$  its fit moves by 8.6 per cent of its length when the reference levels change. Along its path the columns of  $A$  enter at values of  $\lambda$  spread over a factor of 2.8, and it admits its first column of a null factor ( $C$  or  $D$ ) at a residual sum of squares of 112.0, a worse fit than the 110.7 that the group lasso reaches with both null factors still excluded ([Figure 30.3.1](#)).

```
import numpy as np

rng = np.random.default_rng(3003)
n = 120
levels = {"A": 4, "B": 3, "C": 5, "D": 3}
factors = {f: rng.integers(0, k, size=n) for f, k in levels.items()}
```

```

numeric = rng.normal(size=(n, 3))
effect_A = np.array([0.0, 1.0, -0.5, 0.8])
effect_B = np.array([0.0, 0.3, -0.3])
y = effect_A[factors["A"]] + effect_B[factors["B"]] + 0.6 * numeric[:, 0] + rng.
    normal(size=n)
y -= y.mean()

def indicator_columns(codes, k, reference=0):
    """Indicator columns for the levels other than the reference level."""
    return np.column_stack([(codes == l).astype(float) for l in range(k) if l !=
        reference])

def orthonormal_group(Z):
    """Centre the columns, then an orthonormal basis of their span (Z = Q R)."""
    Z = Z - Z.mean(axis=0)
    Q, _ = np.linalg.qr(Z)
    return Q

def build(reference=0):
    blocks = [orthonormal_group(indicator_columns(factors[f], k, reference))
        for f, k in levels.items()]
    blocks += [orthonormal_group(numeric[:, [j]]) for j in range(3)]
    return blocks

def group_lasso(blocks, y, lam, b=None, tol=1e-12, max_sweeps=100000):
    """(1/2)||y - sum_g X_g b_g||^2 + lam sum_g sqrt(p_g) ||b_g||, with X_g'X_g = I.
    """
    b = [np.zeros(Xg.shape[1]) for Xg in blocks] if b is None else [v.copy() for v
        in b]
    r = y - sum(Xg @ bg for Xg, bg in zip(blocks, b))
    for _ in range(max_sweeps):
        largest = 0.0
        for g, Xg in enumerate(blocks):
            s = Xg.T @ r + b[g] # block least squares on the
            partial residual
            w = np.sqrt(Xg.shape[1])
            norm_s = np.linalg.norm(s)
            new = max(0.0, 1 - lam * w / norm_s) * s if norm_s > 0 else 0 * s
            r += Xg @ (b[g] - new)
            largest = max(largest, np.max(np.abs(new - b[g])))
            b[g] = new
        if largest < tol:
            return b
    raise RuntimeError("block coordinate descent did not converge")

blocks = build()
names = list(levels) + ["x1", "x2", "x3"]
lam = 2.0
b = group_lasso(blocks, y, lam)
for name, Xg, bg in zip(names, blocks, b):
    print(f"{name}: {Xg.shape[1]} columns, ||X_g b_g|| = {np.linalg.norm(Xg @ bg):.3
        f}")

```

[Run this code online](#)

The **sparse group lasso** adds an  $\ell_1$  term so that a selected group can have zero coefficients (Simon, Friedman, Hastie and Tibshirani, 2013).

### 30.3.3 Ordered coefficients: the fused lasso

When coefficients have a natural order (levels of an ordered factor, successive times), neighbours are expected to be similar. The **fused lasso** of Tibshirani, Saunders, Rosset, Zhu and Knight (2005) penalizes their differences:

$$Q(\mathbf{b}) = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda_1 \sum_{j=1}^p |b_j| + \lambda_2 \sum_{j=2}^p |b_j - b_{j-1}|. \quad (30.3.4)$$

It produces exact zeros and exact ties  $\hat{\beta}_j = \hat{\beta}_{j-1}$ , so the coefficient sequence is piecewise constant. The penalty is convex but not separable, so coordinate descent can stall at a tie. For an ordered factor, fusion merges adjacent levels the data cannot distinguish (Gertheiss and Tutz, 2010). [Exercise \(B1\)](#) solves the smallest case.

### 30.3.4 Exercises

#### 30.3.4.1 A. Check your understanding

##### Exercise A1.

Under orthonormal groups and  $\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$ , show that  $P(\hat{\beta}_g \neq \mathbf{0}) = P\{\chi^2(p_g) > \lambda^2 w_g^2 / \sigma^2\}$ . With  $w_g = 1$ , compare these probabilities for  $p_g = 1$  and  $p_g = 10$  when  $\lambda = 2\sigma$ . What do the weights  $\sqrt{p_g}$  achieve, and what do they not achieve?

*Solution.* By [Eq. 30.3.3](#),  $\hat{\beta}_g \neq \mathbf{0}$  iff  $\|\mathbf{z}_g\| > \lambda w_g$ . Here  $\mathbf{z}_g = \mathbf{X}_g^\top \mathbf{y} \sim N_{p_g}(\mathbf{0}, \sigma^2 \mathbf{I})$ , so  $\|\mathbf{z}_g\|^2 / \sigma^2 \sim \chi^2(p_g)$ . With  $w_g = 1$  and  $\lambda = 2\sigma$ , the probabilities are  $P\{\chi^2(1) > 4\} = 0.046$  and  $P\{\chi^2(10) > 4\} = 0.947$ , so a large null group almost always enters. With  $w_g = \sqrt{p_g}$  the threshold is  $4p_g$ , four times the mean of  $\chi^2(p_g)$ . This removes the gross imbalance, but it does not equalize the probabilities: they are 0.046 and  $P\{\chi^2(10) > 40\} = 1.7 \times 10^{-5}$ . Equal null entry probabilities would need thresholds at a common quantile of  $\chi^2(p_g)$ .

#### 30.3.4.2 B. Practice

##### Exercise B1.

Solve the fused lasso [Eq. 30.3.4](#) with  $\mathbf{X} = \mathbf{I}_2$ ,  $\lambda_1 = 0$  and  $\lambda_2 = \lambda$ . Show that  $\hat{\beta}_1 = \hat{\beta}_2 = \frac{1}{2}(y_1 + y_2)$  if  $|y_1 - y_2| \leq 2\lambda$ . Otherwise, each estimate moves a distance  $\lambda$  from its observation towards the other.

*Solution.* The penalty is  $\lambda|b_1 - b_2| = \lambda|\mathbf{a}^\top \mathbf{b}|$  with  $\mathbf{a} = (1, -1)^\top$ . Its subgradients are  $\lambda s \mathbf{a}$  with  $s \in \partial|\cdot|(b_1 - b_2)$ . By Lemma 30.1.2,  $y_1 - b_1 = \lambda s$  and  $y_2 - b_2 = -\lambda s$ . If  $b_1 = b_2 = b$ , adding the equations gives  $b = \frac{1}{2}(y_1 + y_2)$ , and then  $s = (y_1 - y_2)/(2\lambda)$  must lie in  $[-1, 1]$ . If  $|y_1 - y_2| > 2\lambda$ , then  $b_1 \neq b_2$ ,  $s = \text{sign}(b_1 - b_2)$ , and  $b_1 = y_1 - \lambda s$ ,  $b_2 = y_2 + \lambda s$ . This is consistent when  $s = \text{sign}(y_1 - y_2)$ , since then  $b_1 - b_2 = y_1 - y_2 - 2\lambda s$  has the sign of  $s$ .

**Exercise B2.**

Under an orthonormal design, compare the group lasso (one group of size 3,  $w = \sqrt{3}$ ) with the lasso on the same three columns, for  $\mathbf{z} = (2, 0.3, -0.2)^\top$  and  $\lambda = 1$ . Which coefficients are zero in each? Show that, under an orthonormal design, the group lasso never sets a single coordinate of a selected group to zero unless the corresponding  $z_j = 0$ .

**30.3.4.3 C. Going deeper****Exercise C1.**

Drop the assumption  $\mathbf{X}_g^\top \mathbf{X}_g = \mathbf{I}$ . Show that a nonzero solution of the block problem  $\min_{\mathbf{b}} \frac{1}{2} \|\mathbf{r} - \mathbf{X}_g \mathbf{b}\|^2 + c \|\mathbf{b}\|$  is  $\mathbf{b} = (\mathbf{X}_g^\top \mathbf{X}_g + (c/t)\mathbf{I})^{-1} \mathbf{X}_g^\top \mathbf{r}$ , where  $t = \|\mathbf{b}\|$  is the root of a one-dimensional equation. Show that this equation has a unique root when  $\|\mathbf{X}_g^\top \mathbf{r}\| > c$ , so the block update is a ridge regression (Theorem 27.2.2) with a data-dependent parameter.